DEPI | Digital Egypt Pioneers Initiative

Capstone documentation · Milestone M1 · 16 October 2026

Retail Intelligence & Market Basket Analysis Platform

Planning Documents

Project Planning & Management · Literature Review · Requirements Gathering

Track	Data Scientist
Project type	Capstone — End-to-End Business Analytics Decision Platform
Supervisor / Lecturer	Sara Ashraf
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Repository	https://github.com/SherifAzb/End-to-End-Business-Analytics-Decision-Platform
Dataset	https://www.kaggle.com/datasets/tawfiqakram/retail-data

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Roles

Member	Role
member1	R1 — Lead + Communication
member2	R2 — Data Engineer
member3	R3 — Cleaning + EDA
member4	R4 — ML + Evaluation
member5	R5 — Dashboard + Deployment

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Part 1 | Project Planning & Management

Proposal, project plan, roles, risks and KPIs.

1.1 Project Proposal

1.1.1 Overview

The project builds a decision-support platform for a supermarket-style retailer. It turns raw sales transactions into three connected insights and delivers them through an interactive dashboard and a table of actionable recommendations:

- Customer segmentation (unsupervised learning): group customers by shopping behaviour (how often they come, how much they spend, how recently they bought, how much they depend on discounts).
- Market basket analysis: find the products that are bought together (for example, customers who buy bread often buy milk), independently of the customer segments.
- Recommender system: suggest the products a customer is most likely to add to the current basket, or to buy next.

1.1.2 Problem statement

The retailer owns thousands of purchase records but cannot answer four practical questions: what do customers buy together, who are the most valuable customers, which products should be recommended to each customer, and how can the average order value be increased. Decisions about shelf layout, bundles and campaigns are therefore taken without evidence.

1.1.3 Objectives

ID	Objective (SMART)	Target date
O1	Build a documented, reproducible data pipeline (SQL schema, cleaning, validation) that produces an analysis-ready dataset and an issues log.	6 Nov 2026
O2	Segment customers with clustering, evaluate the result (Silhouette and bootstrap stability) and profile every segment in business language.	16 Nov 2026
O3	Discover product associations with FP-Growth and report support, confidence and lift for every rule.	16 Nov 2026
O4	Build a recommender and compare it with a popularity baseline on a time-based hold-out period.	25 Nov 2026
O5	Deliver an interactive dashboard that answers all four groups of stakeholder questions.	27 Nov 2026
O6	Deliver a recommendations table (priority, owner, metric to monitor), final report, presentation and infographic.	4 Dec 2026

1.1.4 Scope

In scope	Out of scope
• Data ingestion from CSV into a relational (SQL) database • Data cleaning, validation and a documented issues log • Exploratory data analysis and business KPIs • Customer clustering, association rules and a recommender • Interactive dashboard and recommendations table • Testing, documentation and GitHub repository • Optional: MLflow tracking, FastAPI, Docker, GitHub Actions, cloud deployment	• Real-time streaming or point-of-sale integration • Running live A/B experiments in stores • Price optimisation and demand forecasting • Mobile application • Personal identifying data (names, phone numbers, addresses)

1.1.5 Proposed solution

Raw CSV files are loaded into SQL tables, cleaned and validated, and analysed. The three analytical modules (segmentation, association rules, recommender) read the same cleaned data. Their outputs feed one dashboard and one recommendations table. Every cleaning decision is logged so that the numbers can be traced back to the source.

1.1.6 Data

The platform works on retail transaction data with four related tables:

Table	Main fields
transactions	line id, transaction (invoice) id, customer id, date, product id, product name, category, quantity, unit price, discount, store, payment method, channel
customers	customer id, city, age group, gender, birth year, join date, loyalty member, preferred payment
products	product id, name, category, brand, unit, cost price, list price, perishable flag
stores	store id, name, city, area, region, size class, opening date

Data source and provenance

Dataset name	Retail_Data
Publisher	tawfiqakram (Kaggle)
Link	https://www.kaggle.com/datasets/tawfiqakram/retail-data
Licence	To be confirmed — no licence is stated on the dataset page; record it here and in the README once checked
Date accessed	16 October 2026

Current working dataset size: about 0.8 million transaction lines, 25 stores and 7,000 customers. This is updated in the project plan if the dataset changes.

1.1.7 Technology stack (planned)

- Language and analysis: Python, pandas, NumPy, scikit-learn, mlxtend (FP-Growth), LightGBM, matplotlib, Jupyter notebooks
- Data: SQL database (SQLite or PostgreSQL, to be confirmed in the design phase), CSV and Parquet files
- Dashboard: an interactive dashboard tool to be confirmed with the supervisor
- Collaboration: GitHub (branches, pull requests, issues)
- Optional: MLflow, FastAPI, Docker, GitHub Actions

1.1.8 Methodology

The team follows CRISP-DM (business understanding, data understanding, data preparation, modelling, evaluation, deployment) in short weekly iterations. Work is tracked on a task board, every change goes through a pull request, and a short weekly meeting reviews progress against the milestones.

Starting point: a baseline prototype exists (cleaning pipeline, segmentation, association rules and recommender experiments). During the implementation phase the team reviews, understands, extends and validates it, and documents every decision.

1.2 Project Plan

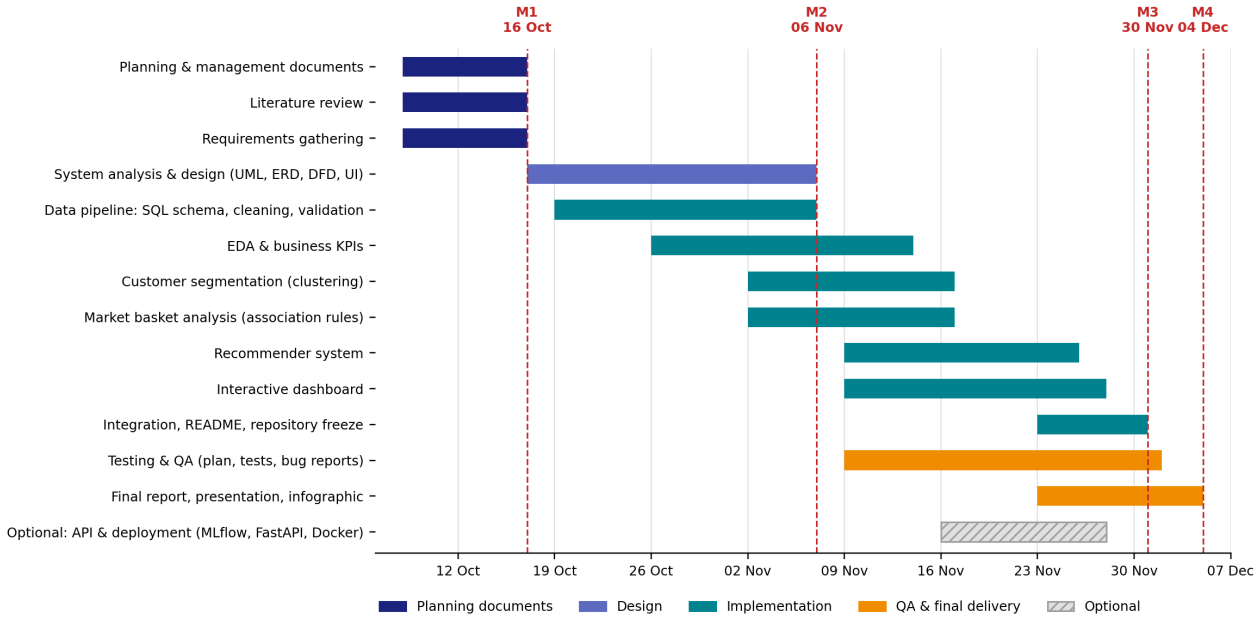
1.2.1 Phases and dates

Dates follow the DEPI documentation schedule (regular DEPI column).

Phase	Main activities	Start	End
A. Planning documents	Planning & management, literature review, requirements gathering	8 Oct	16 Oct
B. System analysis & design	Use cases, ERD, schemas, DFD, UML diagrams, architecture, UI wireframes	17 Oct	6 Nov
C1. Data pipeline	SQL schema, loading, cleaning, validation, issues log	19 Oct	6 Nov
C2. EDA & KPIs	Descriptive analysis, trends, business KPIs	26 Oct	13 Nov
C3. Segmentation and market basket	Clustering, profiling, FP-Growth rules	2 Nov	16 Nov
C4. Recommender	Baselines, hybrid model, time-based evaluation	9 Nov	25 Nov

Phase	Main activities	Start	End
C5. Dashboard	Interactive pages for KPIs, segments, rules, recommendations	9 Nov	27 Nov
C6. Integration and freeze	README, repository clean-up, final source code	23 Nov	30 Nov
D1. Testing & QA	Test plan, test cases, automated tests, bug reports	9 Nov	1 Dec
D2. Final delivery	Final report, presentation, infographic, user manual	23 Nov	4 Dec
E. Optional deployment	API, containers, CI/CD, hosting	16 Nov	27 Nov

1.2.2 Gantt chart



Weeks end on Fridays; M1–M4 are the DEPI deadlines.

1.2.3 Milestones

#	Date	Milestone	Exit criteria
M1	16 Oct 2026	Planning documents submitted	Planning & management, literature review and requirements gathering are on GitHub
M2	6 Nov 2026	System analysis & design submitted; data pipeline ready	Design diagrams complete; cleaned dataset produced by one script; issues log complete
M3	30 Nov 2026	Implementation complete	Models evaluated; dashboard working; README lets another person run everything; source code frozen
M4	4 Dec 2026	Final presentation, testing and reports	Test report, user manual, technical documentation, presentation and infographic submitted

1.2.4 Deliverables

DEPI item	Deliverable	Due	Lead role
Project Planning & Management	This part (proposal, plan, roles, risks, KPIs)	16 Oct	R1
Literature Review	Review of similar solutions and comparison with this project	16 Oct	R1 (with all)
Requirements Gathering	Stakeholders, user stories, use cases, functional and non-functional requirements	16 Oct	R1
System Analysis & Design	Use case, ERD, schemas, DFD, sequence, activity, state and class diagrams, architecture, wireframes	6 Nov	R2 / R5
Implementation	Source code, notebooks, SQL scripts, README, dashboard	30 Nov	R2, R3, R4, R5
Testing & QA	Test plan, test cases, automated tests, bug reports	4 Dec	Shared
Final Presentation & Reports	User manual, technical documentation, presentation, infographic, optional demo video	4 Dec	R1

1.2.5 Resource allocation

Resource	Allocation	Notes
People	5 team members, one role each (see Section 1.3)	Planning assumption: about 8–10 hours per member per week, to be confirmed by the team
Hardware	Team laptops; one-processor, 8 GB RAM is enough for the working dataset	Large files are processed in chunks or on samples during development
Software	Python, Jupyter, Git/GitHub, SQL database, dashboard tool	All free or open-source
Data	Working dataset (CSV) and data dictionary	Source and licence must be recorded in the repository
Budget	None required	Optional cloud deployment only on free tiers

1.3 Task Assignment & Roles

Five roles cover the nine CRISP-DM phases. Each role has a backup who understands the basics of that work, and every pull request is reviewed by a second person.

Role	Responsibilities	Main deliverables	Backup	Reviewer	Assigned to
R1 Lead + Communication	Business case, KPIs, coordination, task board, reports, recommendations table, presentation	Business case, requirements, final report, presentation, infographic	R5	R4	member1
R2 Data Engineer	SQL database, loading scripts, KPI queries, clean data layer, aggregation tables, reproducibility	SQL scripts, ERD and schema, data pipeline, run-from-scratch test	R3	R3	member2
R3 Cleaning + EDA	Data profiling, cleaning rules, issues log, EDA notebook, derived columns	Cleaning notebook, before/after report, EDA notebook	R2	R2	member3
R4 ML + Evaluation	RFM/behavioural features, clustering, association rules, recommender, evaluation	Segmentation, rules and recommender notebooks, evaluation section	R3	R3	member4
R5 Dashboard + Deployment	Dashboard design and build, number checks against SQL, optional API and deployment	Dashboard, wireframes, optional API/Docker	R1	R2	member5

1.3.1 RACI matrix

R = responsible, A = accountable, C = consulted, I = informed.

Deliverable	R1	R2	R3	R4	R5
Business case and KPIs	A/R	C	C	C	I
Requirements and use cases	A/R	C	C	C	C
SQL schema and ERD	C	A/R	C	I	C
Cleaning pipeline and issues log	I	C	A/R	C	I
EDA notebook	C	C	A/R	C	I
Segmentation and association rules	C	I	C	A/R	I
Recommender and evaluation	C	C	C	A/R	C
Dashboard	C	C	I	C	A/R
Testing and bug reports	C	R	R	R	A/R
Final report and presentation	A/R	C	C	C	C
Repository and README	A	R	R	R	R

1.3.2 Working agreements

- Meetings: one 45-minute meeting per week (done, blockers, next week, milestone check).

- Git: protected main branch; feature branches named feat/rN-topic; every pull request reviewed by the person listed above.
- Data contract: table and column names are agreed once and changed only with the whole team's approval.
- Documentation: every decision (what was dropped, filled or capped, and why) is written in the issues log.
- Cross-training: every member must be able to explain the whole project, because the discussion can address anyone.

1.4 Risk Assessment & Mitigation Plan

Probability (P) and impact (I) are scored 1 (low) to 3 (high). Score = P × I. Scores of 6 or more are reviewed at every weekly meeting.

ID	Risk	P	I	Score	Mitigation	Contingency	Owner
R01	Data cleaning takes longer than planned and blocks the modelling work	3	3	9	Start profiling in week 1; modelling uses a sample until clean data is ready	Reduce cleaning scope to critical issues first; document the rest	R3
R02	Dataset source or licence is not documented correctly	2	3	6	Record source, link, licence and access date in the README and in the proposal	Replace the dataset with a documented public one	R1
R03	Data leakage makes model results look better than they are	2	3	6	Time-based split; features built only from the past; peer review of every feature	Re-run the evaluation after removing suspicious features	R4
R04	Team members have limited experience with the tools	3	2	6	Learning plan per role in week 1; pair programming; backup role per task	Re-assign tasks; ask the supervisor for a short session	R1
R05	Merge conflicts and lost work in Git	2	2	4	Feature branches, small pull requests, daily commits	Restore from history; agree one integration owner	R2
R06	Large file makes notebooks slow or crash on laptops	2	2	4	Work on samples in development; read data in chunks; use Parquet	Use a smaller sample for the demo with a note in the report	R2
R07	Scope creep (extra models, extra dashboard pages)	2	3	6	Frozen scope in Section 1.1.4; optional items start only after M3 criteria are met	Cut optional items first	R1
R08	Results are misinterpreted (correlation presented as cause)	2	2	4	Every insight states its evidence and limits; recommend tests before large changes	Add limitations section to the report	R4
R09	Dashboard numbers differ from the SQL or notebook results	2	3	6	Single KPI definition document; numbers checked against SQL in week 6	Fix definitions; re-publish the dashboard	R5
R10	A team member is unavailable for a period	2	2	4	Backup roles; work documented in the repository	Redistribute tasks at the weekly meeting	R1
R11	Deadline pressure before M2, M3 and M4	3	2	6	Plan finishes one week before each DEPI date; checklist reviews	Prioritise must-have items from the requirements	R1
R12	Personal data appears in the repository	1	3	3	Use anonymised identifiers only; no names, phones or addresses; review before every push	Remove from history and rotate any exposed data	R2

1.4.1 Monitoring process

- Risks are reviewed in the weekly meeting and the register is updated in the repository.
- A risk whose score reaches 6 gets a named action and a due date.
- New risks found during the project are added with an owner.

1.5 Key Performance Indicators (KPIs)

KPIs are grouped into project delivery, data quality, analytical quality and system quality. Targets for the analytical KPIs are relative to baselines, so they stay meaningful if the dataset changes.

ID	KPI	Measure	Target	Frequency	Owner
K01	Milestone on-time delivery	Milestones delivered by the DEPI date ÷ total milestones	100% (4 of 4)	Each milestone	R1
K02	Task completion rate	Tasks completed in the week ÷ tasks planned	≥ 85%	Weekly	R1
K03	Documentation completeness	DEPI items submitted ÷ DEPI items required	100%	Each milestone	R1
K04	Pull-request review coverage	Pull requests merged after review ÷ all merged	100%	Weekly	R2
K05	Key completeness after cleaning	Rows with valid transaction and product ids ÷ all rows (true guest purchases reported separately)	≥ 99%	Per pipeline run	R3
K06	Duplicate rows after cleaning	Number of exact duplicate rows	0	Per pipeline run	R3
K07	Issues-log coverage	Issue types with a documented decision ÷ issue types found	100%	Per pipeline run	R3
K08	Segment stability	Adjusted Rand Index between clusterings on bootstrap samples	≥ 0.80	Per model version	R4
K09	Segment interpretability	Segments with a name, a profile and a recommended action ÷ all segments	100%	Per model version	R4
K10	Rule quality	Rules reported with support, confidence and lift; rules with lift ≥ 2 reviewed manually	≥ 10 reviewed	Per model version	R4
K11	Recommender accuracy (Hit@10)	Hit@10 on a time-based hold-out ÷ Hit@10 of the popularity baseline	≥ 1.25×	Per model version	R4
K12	Recommender accuracy (Hit@1)	Hit@1 on the same hold-out ÷ baseline	≥ 2×	Per model version	R4
K13	Dashboard load time (response time)	Time to render each page with filters applied	≤ 5 s	Before M3	R5
K14	Recommendation response time	Time to return the top-5 products for one basket (local)	≤ 1 s	Before M3	R4
K15	Business questions answered (adoption proxy)	Questions answered by the dashboard ÷ stakeholder questions	100%	Before M3	R5
K16	Reproducibility	Pipeline runs end-to-end from the README on a clean machine	Pass	Before M3 and M4	R2
K17	Test pass rate	Passed test cases ÷ executed test cases	≥ 90%	Before M4	All
K18	Actionable recommendations	Recommendations with priority, owner and metric	≥ 5	Before M4	R1

If the work is deployed, an availability target of at least 95% during demonstration days is added to the KPI list.

1.6 Assumptions, dependencies and approval

- The team can dedicate about 8–10 hours per member per week.
- The supervisor confirms the dataset, the dashboard tool and the format of the final deliverables.
- All documents are uploaded to GitHub on or before the DEPI deadlines.
- Optional deployment items are attempted only after the M3 criteria are met.

Role	Name	Signature	Date
Team lead	member1		
Supervisor / lecturer	Sara Ashraf		

Part 2 | Literature Review

Existing methods, datasets, competitions and commercial solutions compared with this project.

2.1 Purpose and method

This review looks at existing methods, datasets, competitions and commercial solutions that address the same problems as this project: segmenting retail customers, finding products that are bought together, and recommending products. For each one we note its strengths and weaknesses, compare it with our project, and decide what to learn from it.

2.1.1 How the review was done

Sources searched: foundational papers on frequent-pattern mining, customer segmentation, collaborative filtering and gradient-boosted ranking; the UCI Machine Learning Repository; the Kaggle Instacart competition; official AWS documentation; public blog posts by practitioners.

Selection rule: a source is included if it addresses market basket analysis, customer segmentation or product recommendation on retail or grocery transactions.

Evidence rule: strengths and weaknesses marked as stated by the source come from the source itself; those marked our assessment are our own judgement and should be read as such.

Date of access: web sources were accessed on 16 October 2026.

2.2 Background concepts

2.2.1 Market basket analysis and association rules

Association rules describe items that tend to appear in the same transaction. A rule “A → B” is described by support (how often A and B appear together), confidence (how often B appears when A is present) and lift (how much more often they appear together than if they were independent; a lift above 1 means a positive association). The classic Apriori algorithm generates candidate itemsets level by level (Agrawal & Srikant, 1994). FP-Growth avoids candidate generation by compressing the data into a prefix tree and reports being about an order of magnitude faster than Apriori in its authors' tests (Han, Pei & Yin, 2000).

2.2.2 Customer segmentation

RFM analysis describes each customer by recency, frequency and monetary value of purchases. Clustering algorithms such as k-means group customers with similar profiles; k-means++ gives a more careful way to choose the starting centres (Arthur & Vassilvitskii, 2007). The Silhouette coefficient measures how well each customer fits its own cluster compared with the nearest other cluster (Rousseeuw, 1987).

2.2.3 Recommender systems

Item-based collaborative filtering recommends items that are similar to what the user already has, where similarity is computed from co-occurrence in user behaviour (Sarwar et al., 2001). Learning-to-rank methods such as LambdaMART re-order a short list of candidates using many features (Burges, 2010), and gradient-boosting libraries such as LightGBM implement this efficiently (Ke et al., 2017).

2.3 Review of similar solutions and projects

#	Solution / source	What it does	Strengths	Weaknesses / limitations	Relevance to our project
A	Apriori — Agrawal & Srikant (1994)	Foundational algorithm for mining association rules in large transaction databases.	Rules are easy to read and explain to business users; widely taught and implemented.	Level-by-level candidate generation becomes expensive when there are many frequent items (our assessment; the FP-Growth authors make the same point).	We understand its measures (support, confidence) but use a faster algorithm.
B	FP-Growth — Han, Pei & Yin (2000)	Mines frequent itemsets from a compressed prefix tree, without generating candidates.	Reported by its authors to be about ten times faster than Apriori; scales to large datasets.	Still returns many rules at low support, so rules must be filtered by lift and checked for meaning (our assessment).	Adopted for our market basket module, with lift filtering.
C	RFM segmentation case study — Chen, Sain & Guo (2012)	Segments customers of an online retailer using the RFM model and data mining techniques.	Simple, interpretable; gives marketing a clear way to rank customers.	Uses only three behaviour variables, so it cannot separate, for example, discount-driven customers from others (our assessment); based on an online, non-store retailer.	Adopted as a baseline; we add basket size, discount use, visit gaps and customer tenure.
D	Instacart Market Basket Analysis — Kaggle competition (2017) on a public grocery-orders dataset	Predict which previously purchased products will appear in a customer's next order. The data covers more than 3 million orders from over 200,000 users.	Large, real, anonymised data; strong feature-engineering and gradient-boosting solutions; one top solution is described by NVIDIA as using gradient-boosted trees and a special treatment of the F1 metric.	Focuses on reorder prediction for existing customers; does not produce customer segments, bundle rules or a recommendations table for managers (our assessment).	Adopted: time-aware evaluation and gradient-boosting for re-ranking. Candidate dataset for validation.
E	Online Retail II — UCI Machine Learning Repository (Chen, 2019)	Public dataset of two years of transactions of a UK-based online retailer; about 1.07 million records; CC BY 4.0.	Real transactions; free licence; widely used for RFM, clustering and basket analysis.	Online gift-ware retailer with many wholesale customers, so behaviour differs from a supermarket; contains missing values and cancellation invoices that need cleaning.	Candidate dataset; shows the cleaning issues we also handle.
F	Practitioner blog projects on Instacart data (BigML, 2018; NYC Data Science Academy, 2018)	Prepare grocery orders for association-rule mining; build an interactive dashboard for promotions.	Accessible; show the full path from data to a first insight.	From our reading they stop at descriptive analysis or rule discovery and do not combine segmentation, rules, a recommender and a documented evaluation (our assessment).	We follow the same data preparation idea (one row per order) but extend the scope.
G	Item-based collaborative filtering — Sarwar et al. (2001)	Recommends items similar to those a user already has, based on co-occurrence.	Scalable and easy to explain (“customers who bought X also bought Y”).	Weak when there is little history for a customer or product (cold start); ignores context such as season (our assessment).	Adopted as one scoring component (cosine similarity) inside our hybrid model.
H	Learning to rank with gradient boosting — Burges (2010); Ke et al. (2017)	Re-orders a candidate list using many features.	High accuracy when good features exist; fast to train.	Needs labelled data and careful time-based validation to avoid leakage (our assessment).	Adopted as the final re-ranking step.
I	Amazon Personalize — AWS (documentation accessed 16 Oct 2026)	Fully managed service that generates item recommendations from user interaction data; includes retail-oriented recommenders such as “Frequently Bought Together”.	Managed infrastructure, real-time recommendations, integration with marketing and A/B testing tools (stated by AWS).	A paid commercial service; the models are managed by the provider so the team cannot inspect or change them (our assessment); not designed to explain results to a student audience.	We build and explain each step ourselves; the service shows what a production system offers.

2.4 Comparison with this project

The table compares what each approach is mainly designed to do. “Not the focus” means the source does not set out to do this; it does not mean the work is poor.

Criterion	This project	Apriori / FP-Growth rules	RFM segmentation study	Instacart competition solutions	Amazon Personalize
Customer segmentation	Yes — clustering on behaviour features	Not the focus	Yes — RFM-based	Not the focus	Supports segment-based campaigns (stated by AWS)
Products bought together	Yes — FP-Growth with lift	Yes — core purpose	Not the focus	Indirect (reorder prediction)	Yes — “Frequently Bought Together”
Recommender	Yes — hybrid with re-ranking	No	No	Yes — reorder prediction	Yes
Documented data cleaning and issues log	Yes — required deliverable	Not the focus	Not the focus	Not the focus	Not the focus
Time-based evaluation	Yes — hold-out by date	Not the focus	Not the focus	Yes — train/test by orders over time	Provider-side
Business recommendations (priority, owner, metric)	Yes — required deliverable	Not the focus	Partly — marketing use	Not the focus	Not the focus
Explainable to non-technical users	Yes — every recommendation shows a reason	Yes	Yes	Limited (complex features)	Limited (managed models)
Reproducible code in the repository	Yes	Depends on the library	Not stated	Public notebooks by participants	No — managed service
Cost	Free (open-source tools)	Free	Free	Free	Paid usage-based service

2.5 What we adopt and how we differ

2.5.1 Lessons we apply

- Use lift, not only confidence: a rule can have high confidence only because the product is popular. Lift corrects for this (Section 2.2.1).
- Prefer FP-Growth to Apriori for speed on large transaction tables.
- Go beyond RFM: add discount use, basket size, visit gaps and tenure to separate customer types that RFM mixes together.
- Evaluate over time: train on the past and test on a later period, as the Instacart competition does with prior and later orders.
- Use simple baselines first: popularity and item co-occurrence, then a gradient-boosting re-ranker.
- Check the data source: public datasets such as Online Retail II document their contents and licence, and we do the same for ours.

2.5.2 What is different in our project

- Three connected modules (segments, rules, recommender) are built on one cleaned dataset and delivered in one dashboard.
- The cleaning process is a deliverable: every issue and decision is logged with before/after counts.
- Results end in a recommendations table with priority, owner and metric to monitor, which the stakeholder asked for.
- Every recommendation carries a reason a store manager can understand.

2.5.3 Possible improvements suggested by the review

- Validate the pipeline on a second public dataset (for example Online Retail II) to check that it does not depend on one dataset.
- Test recommendations with a small A/B experiment, because association is not causation.
- Add seasonality (for example holidays or Ramadan) as an explicit feature if the data covers it.
- Study cold-start handling for new customers and new products.

2.6 Limitations of this review

- It covers a selection of widely cited methods and public resources, not every study in the field.
- Web pages and blog posts can change; they were accessed on 16 October 2026.
- Statements about weaknesses marked “our assessment” are our own judgement and have not been tested experimentally.

2.7 Lecturer feedback and evaluation

To be completed by the lecturer (DEPI documentation guideline: Feedback & Evaluation, Suggested Improvements, Final Grading Criteria).

Item	Lecturer notes
Feedback & evaluation	
Suggested improvements	
Final grading criteria (marks for documentation, implementation, testing, presentation)	

2.8 References

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References 1, 3–8 are cited from the published literature; the team should open each paper once to confirm page numbers before final submission. Web sources (9–15) were checked on 16 October 2026.

Part 3 | Requirements Gathering

What the platform must do and how well it must do it.

3.1 Introduction

3.1.1 Purpose

This part states what the platform must do and how well it must do it. It is the basis for the design (due 6 November 2026), the implementation, and the test plan.

3.1.2 How the requirements were gathered

Stakeholder conversation: a scenario conversation between the Store Manager (business) and the Data Analyst, in which the manager asks for decisions on shelf layout, bundles, recommendations at checkout and a higher average order value, and asks for recommendations rather than only charts.

Business questions: four question groups (Product, Customer, Recommendation, Business) listed in the project brief.

Data review: inspection of the available tables and their quality problems.

Team workshop: the five team members reviewed the list and agreed priorities (MoSCoW: Must, Should, Could, Won't).

3.1.3 The business questions

Group	Questions
Product	Which products sell most? Which are bought together? Which combinations are strongest? Which bundles are possible?
Customer	Who are the best customers? Who buys a lot? Who stopped buying? Does behaviour differ between segments?
Recommendation	If a customer buys X, what do we suggest? If a customer buys X and Y, what is likely next? Which cross-sell opportunities are best?
Business	Where should products be placed in the store? Which bundles should we create? Which products to suggest at checkout? How do we increase average order value?

3.2 Stakeholder analysis

Stakeholder	Role	Main needs	Influence	Interest	How we involve them
Store / Commercial Manager	Decision maker; owner of the request	Clear answers to the business questions; actions he can take to the teams; trustworthy numbers	High	High	Weekly review of results; sign-off on KPI definitions
Marketing team	Runs campaigns and offers	Customer segments with profiles and suggested actions; bundle ideas	High	High	Segment profile review; campaign recommendations
Merchandising / category team	Chooses assortment and prices	Best and worst products; product combinations; seasonal patterns	Medium	High	Product and rules pages of the dashboard
Store operations	Arranges shelves and checkout	Which products to place together; what to suggest at checkout	Medium	High	Shelf-layout and checkout recommendations
CRM / Loyalty manager	Manages loyalty members and win-back	Lists of lapsed and high-value customers; reasons	Medium	High	Segment tables and customer lists
Data / IT team	Maintains data and systems	Reproducible pipeline; clear data dictionary; no sensitive data in files	Medium	Medium	Documentation and pipeline README
Academic supervisor / lecturer	Evaluates the project	Complete documentation; correct methods; honest reporting of data and limits	High	High	Milestone reviews on the DEPI dates
Project team (5 members)	Builds the platform	Clear scope, achievable plan, shared understanding	High	High	Weekly meeting; GitHub board

Influence/interest guidance: stakeholders with high influence and high interest are consulted before every milestone; the others are informed after.

3.3 User stories

Format: As a <role>, I want <goal>, so that <benefit>. Priority uses MoSCoW.

ID	User story	Acceptance criteria	Priority
US-01	As a store manager, I want to see revenue, number of invoices, average order value and trends on one page, so that I understand business performance quickly.	Dashboard shows these KPIs with date, store and category filters; numbers match the SQL check queries.	Must
US-02	As a merchandising analyst, I want to see top and bottom products and categories by revenue and quantity, so that I can adjust the assortment.	Ranked tables and charts with filters; export to CSV.	Must
US-03	As a marketing manager, I want customers grouped into segments with clear names and profiles, so that I can plan different campaigns.	Each segment has a name, size, share of revenue, average basket value, visit frequency, discount use and one recommended action.	Must
US-04	As a CRM manager, I want a list of customers who stopped buying, so that I can run a win-back campaign.	Customer list for the lapsed segment, downloadable, with last purchase date.	Must
US-05	As a store operations manager, I want to know which products are bought together, so that I can place them near each other.	Rules table with support, confidence and lift; filter by minimum lift; plain-language reading of each rule.	Must
US-06	As a marketing manager, I want bundle ideas based on strong product pairs or triples, so that I can create offers.	At least five bundle candidates with their lift and an estimated share of baskets affected.	Should
US-07	As a cashier or checkout system, I want the top 5 suggested products for the current basket, so that I can offer add-ons.	Given a basket, the system returns 5 products that are not already in it, each with a short reason.	Must
US-08	As a CRM manager, I want suggestions for a specific customer, so that I can personalise messages.	Given a customer id, the system returns suggested products using the customer's history and segment.	Should
US-09	As a store manager, I want a recommendations table with priority, owner and metric to monitor, so that I can take clear actions to the Marketing, Operations and CRM teams.	Table with at least five actions; each action has an owner team, a priority and a metric.	Must

ID	User story	Acceptance criteria	Priority
US-10	As a data analyst, I want to rerun the whole pipeline with one command, so that results can be reproduced.	README steps run from a clean machine and produce the same outputs.	Must
US-11	As a data analyst, I want a log of every cleaning decision with before/after counts, so that nobody doubts the numbers.	Issues log lists each issue, rows affected, decision and reason.	Must
US-12	As a data/IT manager, I want no personal identifying data in the repository, so that privacy is protected.	Only anonymised identifiers appear in files; review checklist signed before each push.	Must
US-13	As a supervisor, I want to see how well the models perform, so that I can judge the quality of the work.	Evaluation section reports metrics, baselines, limitations and the reason for model choice.	Must
US-14	As a store manager, I want to understand seasonal effects, so that I can prepare stock.	Dashboard shows monthly and weekday patterns and any holiday-period uplift found in the data.	Could
US-15	As a developer, I want the recommender available as an API, so that other systems can call it.	POST endpoint returns recommendations; documented in the README.	Could

3.4 Use cases

3.4.1 Actors

- Store / Commercial Manager — views KPIs, segments, rules and recommendations.
- Marketing / CRM user — explores segments and customer lists.
- Store operations / cashier — asks for product suggestions for a basket.
- Data analyst — runs the pipeline, reviews data quality, refreshes results.
- Supervisor — reviews documentation and results.

3.4.2 Use case summary

ID	Use case	Primary actor	Related stories
UC-01	View business KPIs and trends	Store Manager	US-01, US-14
UC-02	Explore products and categories	Merchandising analyst	US-02
UC-03	Explore customer segments	Marketing / CRM	US-03, US-04
UC-04	Explore product associations (rules)	Store operations	US-05, US-06
UC-05	Get recommendations for a basket	Cashier / operations	US-07
UC-06	Get recommendations for a customer	CRM	US-08
UC-07	View and export recommended actions	Store Manager	US-09
UC-08	Run data pipeline and review the issues log	Data analyst	US-10, US-11

3.4.3 Use case descriptions

UC-03 — Explore customer segments

Actor: Marketing / CRM user

Precondition: Data pipeline and segmentation have been run; the dashboard is open.

Main flow:

1. User opens the Segments page.
2. System shows the segments with size, revenue share and profile.
3. User selects a segment.
4. System shows the segment profile, favourite categories and recommended action.
5. User exports the customer list of the segment.

Alternative flows: No segment data available → the system shows a message that the pipeline must be run.

Postcondition: User has the list and the profile needed to plan a campaign.

UC-05 — Get recommendations for a basket

Actor: Cashier / operations user

Precondition: The recommender has been trained and loaded.

Main flow:

1. User enters the products in the current basket (and optionally a customer id).
2. System checks that each product exists.
3. System scores candidate products and ranks them.
4. System returns the top 5 products with a short reason for each.

Alternative flows: Unknown product name → the system suggests the closest known names or reports the product as unknown. Empty basket → recommendations use the customer's history, segment and popularity.

Postcondition: User sees five products not already in the basket.

UC-08 — Run data pipeline and review the issues log

Actor: Data analyst

Precondition: Raw files are in the data folder; dependencies installed.

Main flow:

1. Analyst runs the pipeline steps in order (cleaning, segmentation, rules, recommender).
2. System writes the clean dataset, the issues log and result tables.
3. Analyst opens the issues log and checks before/after counts.

Alternative flows: A step fails → the system stops with a clear error message and the failed step is shown in the log.

Postcondition: Clean data, logs and results are available for the dashboard.

3.5 Functional requirements

Priority: M = must, S = should, C = could.

Module	ID	Requirement	Priority	Traces to
Data ingestion and cleaning	FR-01	The system shall load the raw CSV files (transactions, customers, products, stores) into SQL tables without changing the raw values.	M	US-10
	FR-02	The system shall read all fields as text first so that unusual values are not lost.	S	US-11
	FR-03	The system shall detect and handle duplicates, missing keys, invalid quantities, price outliers, invalid discounts, inconsistent labels and mixed date formats.	M	US-11
	FR-04	The system shall recover missing values from other records of the same transaction or from reference tables where possible, and drop records only when recovery is impossible.	S	US-11
	FR-05	The system shall write an issues log with the issue, rows affected, decision and reason.	M	US-11
	FR-06	The system shall validate foreign keys (customer, product, store) and report orphan records.	M	US-11
EDA and KPIs	FR-07	The system shall compute revenue, number of transactions, average order value, items per basket, discount share and margin.	M	US-01
	FR-08	The system shall provide trend views by month, weekday, store, region and category.	M	US-01, US-14
	FR-09	The system shall rank products and categories by revenue and quantity.	M	US-02
Segmentation	FR-10	The system shall compute behaviour features per customer (recency, frequency, monetary value, average order value, visit gap, discount share, items per basket, tenure).	M	US-03
	FR-11	The system shall cluster customers with at least two configurations and report Silhouette and stability.	M	US-03, US-13

Module	ID	Requirement	Priority	Traces to
	FR-12	The system shall profile each segment and assign a name and a recommended action.	M	US-03
	FR-13	The system shall export the customer list of each segment.	S	US-04
Market basket	FR-14	The system shall mine association rules with FP-Growth using configurable minimum support, confidence and lift.	M	US-05
	FR-15	The system shall display each rule with support, confidence and lift and a plain-language reading.	M	US-05
	FR-16	The system shall list bundle candidates derived from the strongest rules.	S	US-06
Recommender	FR-17	The system shall return the top-N products for a given basket, excluding products already in the basket.	M	US-07
	FR-18	The system shall support recommendations for a given customer id.	S	US-08
	FR-19	The system shall return a short reason with every recommendation.	S	US-07
	FR-20	The system shall be evaluated against a popularity baseline on a time-based hold-out using Hit@K and NDCG.	M	US-13
Dashboard and reporting	FR-21	The system shall provide an interactive dashboard with pages for KPIs, products, segments, rules and recommendations, with date, store, category and segment filters.	M	US-01 to US-06
	FR-22	The system shall provide a recommendations table with priority, owner and metric to monitor.	M	US-09
	FR-23	The system shall let users export tables as CSV files.	S	US-02, US-04
Operations	FR-24	The system shall run end-to-end from documented commands and fixed random seeds.	M	US-10
	FR-25	The system shall log errors with the failing step and a readable message.	S	US-10
	FR-26	The system should expose the recommender through a documented API endpoint.	C	US-15

3.6 Non-functional requirements

ID	Category	Requirement	Measure / verification
NFR-01	Performance	Dashboard pages shall render within 5 seconds with filters applied.	Timed test on a team laptop
NFR-02	Performance	One basket recommendation shall be returned within 1 second (local run).	Timed test on 100 requests
NFR-03	Performance	The full pipeline shall run within 15 minutes on a laptop with 8 GB RAM for the working dataset.	Timed end-to-end run
NFR-04	Scalability	The pipeline shall handle at least 1 million transaction lines by processing in chunks or with efficient formats.	Run on a larger file
NFR-05	Security and privacy	Data shall contain anonymised identifiers only; no personal identifying information shall be stored in the repository.	Pre-push checklist; repository review
NFR-06	Security	Credentials, keys and tokens shall never be committed; configuration uses environment variables.	Repository scan
NFR-07	Security	Inputs (file names, product names, ids) shall be validated before use.	Unit tests with invalid input
NFR-08	Usability	Dashboard labels and recommendations shall use plain business language; each chart has a title and axis labels.	Review by two non-technical users
NFR-09	Usability / accessibility	Colours shall have sufficient contrast and be distinguishable for colour-blind users; text size is readable on a projector.	Visual review
NFR-10	Reliability	Results shall be reproducible: the same input and seed produce the same outputs.	Two runs compared
NFR-11	Reliability	A failed step shall stop the pipeline with a clear message and shall not leave partial outputs unmarked.	Failure-injection test
NFR-12	Maintainability	Code shall be modular, commented, and follow one naming convention; each module has a docstring.	Code review checklist
NFR-13	Portability	The project shall run on Windows and Linux with Python 3.10 or later and a requirements file.	Test on two machines
NFR-14	Explainability	Every segment, rule and recommendation shall be explainable in business terms.	Reviewed with the stakeholder scenario
NFR-15	Data quality	Every cleaning decision shall be documented with counts before and after.	Issues log complete
NFR-16	Compliance / licensing	The source and licence of the data shall be recorded in the README and the proposal.	Documentation check
NFR-17	Availability (if deployed)	The deployed demo shall be available at least 95% of the time on demonstration days.	Uptime check

3.7 Constraints, assumptions and out-of-scope items

3.7.1 Constraints

- Academic project with fixed DEPI deadlines (16 Oct, 6 Nov, 30 Nov, 4 Dec 2026).
- Team of five members with mixed experience.
- Free and open-source tools only; no paid services required.
- All documents are uploaded to GitHub.

3.7.2 Assumptions

- The working dataset has the structure described in the proposal (Section 1.1.6).
- Customers are identified by an anonymised id; a small share of purchases are from unidentified (guest) customers.
- Past behaviour in the data is a reasonable guide to near-future behaviour.
- The supervisor confirms the dataset source and the dashboard tool.

3.7.3 Out of scope

- Real-time integration with point-of-sale systems.
- Live A/B experiments in stores.
- Price optimisation and demand forecasting.
- Handling of personal identifying information.

3.8 Traceability matrix

Each business question is linked to the requirements that answer it.

Business question group	User stories	Functional requirements	Non-functional requirements	Planned test
Product questions	US-02, US-05, US-06	FR-07, FR-09, FR-14, FR-15, FR-16	NFR-01, NFR-14	Rule values match a manual count on a sample
Customer questions	US-03, US-04	FR-10, FR-11, FR-12, FR-13	NFR-10, NFR-14	Segments reproducible; stability ≥ 0.80
Recommendation questions	US-07, US-08	FR-17, FR-18, FR-19, FR-20	NFR-02, NFR-14	Basket never contains a recommended product; Hit@K vs baseline
Business questions	US-01, US-09, US-14	FR-07, FR-08, FR-21, FR-22	NFR-01, NFR-08	Dashboard numbers equal SQL results
Data trust	US-10, US-11, US-12	FR-01 to FR-06, FR-24, FR-25	NFR-03, NFR-05, NFR-10, NFR-15, NFR-16	End-to-end rerun; issues log complete

3.9 Open questions and sign-off

3.9.1 Open questions to confirm with the supervisor

- Is the Kaggle dataset Retail_Data accepted, and what is its licence?
- Which dashboard tool is accepted for the final delivery?
- Is Market Basket Analysis accepted as a machine-learning component next to clustering (the project brief lists regression, classification and clustering)?
- Which deployment option, if any, is expected?

3.9.2 Approval

Role	Name	Signature	Date
Team lead	member1		
Stakeholder representative / supervisor	Sara Ashraf		